Lithology Classification Colab

gg · 9 de abril de 2026

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Lithology Classification from Well Logs — Beginner Session

Dataset: FORCE 2020 — North Sea wells

Target: LITH — Lithology class (Shale, Sandstone, Limestone, ...)

Features: GR · RHOB · NPHI · PEF · RDEP · DTC

Blind well: 16/1-2 (last well — used ONLY for final evaluation)

Workflow

1. Load & inspect data
2. Clean data
3. Visualize (histogram · boxplot · log tracks · class balance)
4. Encode labels & scale features
5. Split train / validation / blind
6. Compare models (LazyPredict style)
7. Best model — detailed results
8. Blind well prediction + confusion matrix

STEP 1 — Colab Setup: Upload CSV

Run this cell first to upload the CSV file.

In [1]:

```
from google.colab import drive
drive.mount('/content/drive')
```

Out [1]:

```
Mounted at /content/drive
```

In [4]:

```
# # ── Upload CSV to Colab ──────────────────────────────────────────
# from google.colab import files

# print('Please upload:  training_file.csv')
# uploaded = files.upload()
# csv_filename = list(uploaded.keys())[0]
# print(f'Uploaded: {csv_filename}')
```

In [6]:

```
# Load the dataset
df = pd.read_csv('/content/drive/MyDrive/SPE/Wells_Lith.csv')

# Look at the first 5 rows to see what we have
df.head()
```

Out [6]:

	WELL	DEPTH_MD	GROUP	FORMATION	CALI	RDEP	RHOB	GR	NPHI	PEF	DTC	LITH
0	15/9-13	494.528	NORDLAND GP.	NaN	19.480835	1.798681	1.884186	80.200851	NaN	20.915468	161.131180	Shale
1	15/9-13	494.680	NORDLAND GP.	NaN	19.468800	1.795641	1.889794	79.262886	NaN	19.383013	160.603470	Shale
2	15/9-13	494.832	NORDLAND GP.	NaN	19.468800	1.800733	1.896523	74.821999	NaN	22.591518	160.173615	Shale
3	15/9-13	494.984	NORDLAND GP.	NaN	19.459282	1.801517	1.891913	72.878922	NaN	32.191910	160.149429	Shale
4	15/9-13	495.136	NORDLAND GP.	NaN	19.453100	1.795299	1.880034	71.729141	NaN	38.495632	160.128342	Shale

📦 STEP 2 — Import Libraries

In [9]:

```
# ── Data & math ──────────────────────────────────────────────────
import numpy as np
import pandas as pd
import io

# ── Plotting ─────────────────────────────────────────────────────
import matplotlib.pyplot as plt
import seaborn as sns

# ── ML: preprocessing ────────────────────────────────────────────
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split

# ── ML: classification models ────────────────────────────────────
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from xgboost import XGBClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB

# ── ML: evaluation ───────────────────────────────────────────────
from sklearn.metrics import (accuracy_score, f1_score,
                             classification_report, confusion_matrix,
                             ConfusionMatrixDisplay)

# ── Settings ─────────────────────────────────────────────────────
import warnings
warnings.filterwarnings('ignore')
plt.rcParams['figure.figsize'] = (12, 5)
plt.rcParams['font.size'] = 11

print('All libraries imported!')
```

Out [9]:

```
All libraries imported!
```

📂 STEP 3 — Load & Inspect the Data

In [10]:

```
# ── Load CSV ──────────────────────────────────────────────────────────
df = pd.read_csv(io.BytesIO(uploaded[csv_filename]))

print(f'Shape: {df.shape}   ({df.shape[0]:,} rows x {df.shape[1]} columns)')
print(f'\nColumns: {df.columns.tolist()}')
display(df.head())
```

Out [10]:

```
Shape: (133198, 12)   (133,198 rows x 12 columns)

Columns: ['WELL', 'DEPTH_MD', 'GROUP', 'FORMATION', 'CALI', 'RDEP', 'RHOB', 'GR', 'NPHI', 'PEF', 'DTC', 'LITH']
```

	WELL	DEPTH_MD	GROUP	FORMATION	CALI	RDEP	RHOB	GR	NPHI	PEF	DTC	LITH
0	15/9-13	494.528	NORDLAND GP.	NaN	19.480835	1.798681	1.884186	80.200851	NaN	20.915468	161.131180	Shale
1	15/9-13	494.680	NORDLAND GP.	NaN	19.468800	1.795641	1.889794	79.262886	NaN	19.383013	160.603470	Shale
2	15/9-13	494.832	NORDLAND GP.	NaN	19.468800	1.800733	1.896523	74.821999	NaN	22.591518	160.173615	Shale
3	15/9-13	494.984	NORDLAND GP.	NaN	19.459282	1.801517	1.891913	72.878922	NaN	32.191910	160.149429	Shale
4	15/9-13	495.136	NORDLAND GP.	NaN	19.453100	1.795299	1.880034	71.729141	NaN	38.495632	160.128342	Shale

In [11]:

```
# ── Class distribution — how many samples per lithology? ──────────────
print('Lithology class counts:')
print(df['LITH'].value_counts())

print('\nWells in dataset:')
print(df['WELL'].value_counts())

print('\nMissing values per column:')
print(df.isnull().sum())
```

Out [11]:

```
Lithology class counts:
LITH
Shale              82390
Sandstone          15794
Limestone          11912
Sandstone/Shale     6573
Chalk               5129
Marl                4968
Halite              4314
Tuff                1498
Anhydrite            374
Dolomite             188
Coal                  58
Name: count, dtype: int64

Wells in dataset:
WELL
15/9-13        18270
15/9-15        17717
16/10-1        17675
15/9-17        17350
16/10-3        15956
16/11-1 ST3    14938
16/2-11 A      10708
16/2-16        10025
16/1-6 A        3623
16/10-5         2765
16/10-2         2437
16/1-2          1734
Name: count, dtype: int64

Missing values per column:
WELL             0
DEPTH_MD         0
GROUP            0
FORMATION    21566
CALI           192
```

```
RDEP          7393
RHOB         25145
GR               0
NPHI         41473
PEF          32358
DTC            563
LITH             0
dtype: int64
```

🧹 STEP 4 — Clean Data

Select the 6 log features + target, drop NaN rows, apply physical range filters. We use **15/9-13** as the blind well — it is separated BEFORE cleaning.

In [12]:

```
# ── Define features and target ──────────────────────────────────
FEATURES = ['GR', 'RHOB', 'NPHI', 'PEF', 'RDEP', 'DTC']
TARGET   = 'LITH'

# ── Separate blind well FIRST (before any processing) ──────────
# Well 15/9-13 is our blind well — never used for training
BLIND_WELL = '15/9-13'
df_blind_raw = df[df['WELL'] == BLIND_WELL].copy()
df_train_raw = df[df['WELL'] != BLIND_WELL].copy()

print(f'Training wells rows : {len(df_train_raw):,}')
print(f'Blind well rows     : {len(df_blind_raw):,}')
```

Out [12]:

```
Training wells rows : 114,928
Blind well rows     : 18,270
```

In [13]:

```
# ── Keep only needed columns, drop NaN, apply range filters ────

def clean_well_data(df_in):
    # Keep only feature + target columns
    df_out = df_in[FEATURES + [TARGET, 'WELL', 'DEPTH_MD']].copy()

    # Drop rows where any feature or target is NaN
    df_out = df_out.dropna(subset=FEATURES + [TARGET])

    # Physical range filters (remove unrealistic sensor readings)
    df_out = df_out[(df_out['GR']   >=   0) & (df_out['GR']   <= 200)]
    df_out = df_out[(df_out['RHOB'] >= 1.5) & (df_out['RHOB'] <= 3.5)]
    df_out = df_out[(df_out['NPHI'] >=-0.1) & (df_out['NPHI'] <= 1.0)]
    df_out = df_out[(df_out['PEF']  >=   0) & (df_out['PEF']  <=  10)]
    df_out = df_out[(df_out['RDEP'] >= 0.1) & (df_out['RDEP'] <=2000)]
    df_out = df_out[(df_out['DTC']  >=  40) & (df_out['DTC']  <= 250)]

    return df_out.reset_index(drop=True)

df_train = clean_well_data(df_train_raw)
df_blind = clean_well_data(df_blind_raw)

print(f'Training rows after cleaning : {len(df_train):,}')
print(f'Blind rows after cleaning    : {len(df_blind):,}')
print('\nTraining class distribution:')
print(df_train[TARGET].value_counts())
```

Out [13]:

```
Training rows after cleaning : 67,858
Blind rows after cleaning    : 11,927

Training class distribution:
LITH
Shale              39696
Limestone           9377
Sandstone           8262
Sandstone/Shale     4145
Marl                3127
```

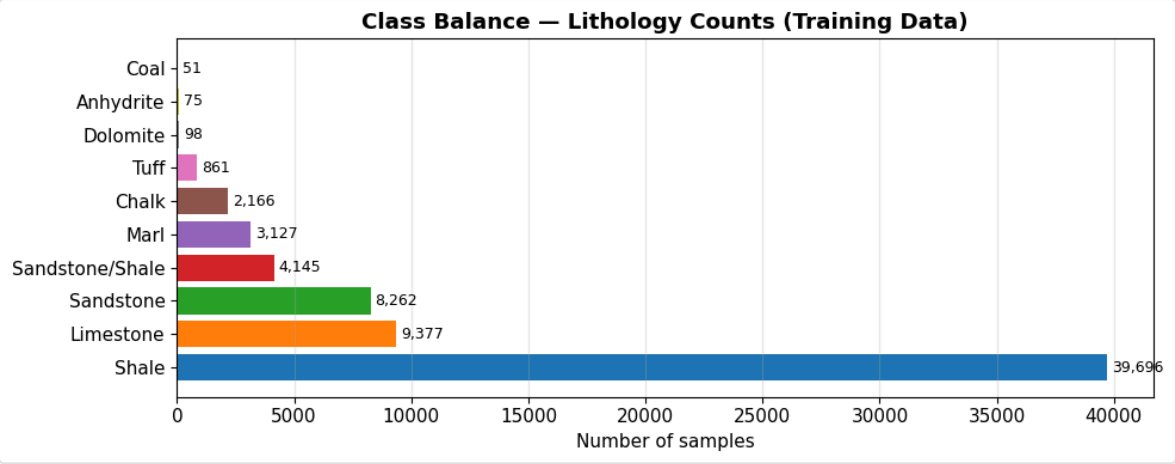
```
Chalk               2166
Tuff                 861
Dolomite              98
Anhydrite             75
Coal                  51
Name: count, dtype: int64
```

📊 STEP 5 — Exploratory Visualization

In [14]:

```
# ── 5.1 Class balance bar chart ──────────────────────────────────────
# Important for classification — imbalanced classes can bias the model

class_counts = df_train[TARGET].value_counts()

fig, ax = plt.subplots(figsize=(10, 4))
bars = ax.barh(class_counts.index, class_counts.values,
               color=plt.cm.tab10.colors[:len(class_counts)])
ax.set_xlabel('Number of samples')
ax.set_title('Class Balance — Lithology Counts (Training Data)', fontweight='bold')

# Add count labels on bars
for bar, count in zip(bars, class_counts.values):
    ax.text(bar.get_width() + 200, bar.get_y() + bar.get_height()/2,
            f'{count:,}', va='center', fontsize=9)

ax.grid(True, axis='x', alpha=0.3)
plt.tight_layout()
plt.show()
print('Notice: Shale dominates the dataset — this is a class imbalance problem')
```

Out [14]:



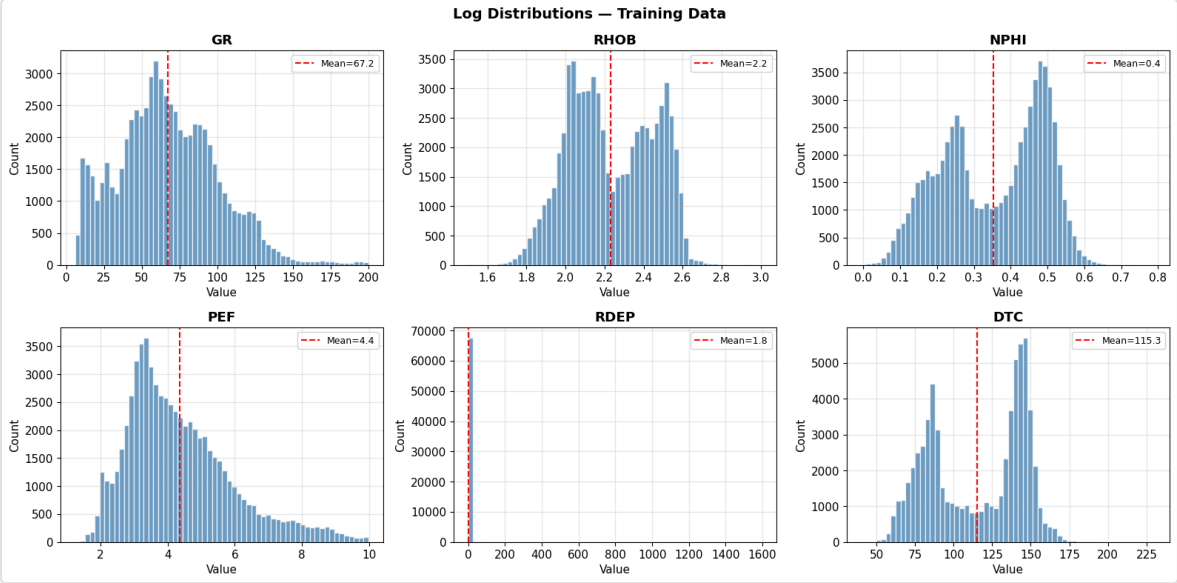

```
Notice: Shale dominates the dataset — this is a class imbalance problem
```

In [15]:

```
# ── 5.2 Histograms — distribution of each log ────────────────────────
fig, axes = plt.subplots(2, 3, figsize=(16, 8))
fig.suptitle('Log Distributions — Training Data', fontsize=14, fontweight='bold')
axes = axes.flatten()

for i, col in enumerate(FEATURES):
    data = df_train[col].dropna()
    axes[i].hist(data, bins=60, color='steelblue', edgecolor='white', alpha=0.8)
    axes[i].axvline(data.mean(), color='red', ls='--', lw=1.5,
                    label=f'Mean={data.mean():.1f}')
    axes[i].set_title(col, fontweight='bold')
    axes[i].set_xlabel('Value')
    axes[i].set_ylabel('Count')
    axes[i].legend(fontsize=9)
    axes[i].grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.show()
```

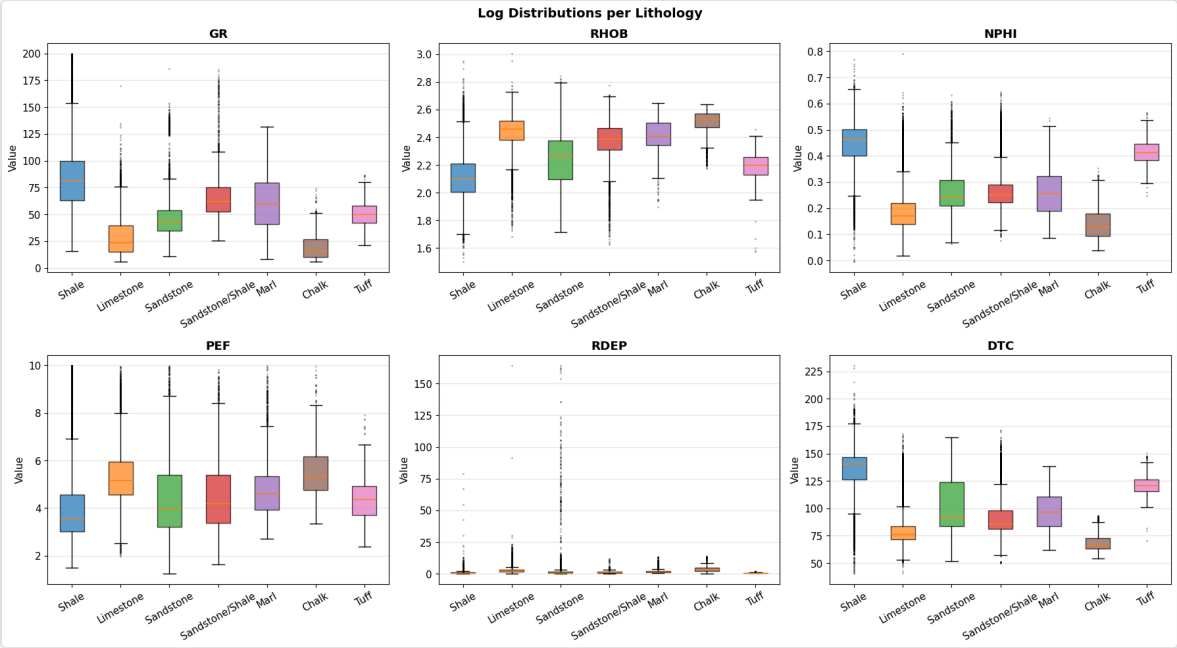
Out [15]:



In [16]:

```
# ── 5.3 Boxplot — log distributions per lithology ──────────────────────
# This is the MOST useful plot for classification:
# It shows how each lithology separates in log space

fig, axes = plt.subplots(2, 3, figsize=(18, 10))
fig.suptitle('Log Distributions per Lithology', fontsize=14, fontweight='bold')
axes = axes.flatten()

# Only plot lithologies with enough samples
main_liths = df_train[TARGET].value_counts()[lambda x: x > 100].index.tolist()
df_main = df_train[df_train[TARGET].isin(main_liths)]

for i, col in enumerate(FEATURES):
    ax = axes[i]
    # Group data by lithology
    data_per_lith = [df_main[df_main[TARGET] == lith][col].dropna().values
                     for lith in main_liths]

    bp = ax.boxplot(data_per_lith, labels=main_liths, patch_artist=True,
                    flierprops=dict(marker='.', markersize=2, alpha=0.3))

    # Color each lithology box differently
    colors = plt.cm.tab10.colors
    for patch, color in zip(bp['boxes'], colors):
        patch.set_facecolor(color)
        patch.set_alpha(0.7)

    ax.set_title(col, fontweight='bold')
    ax.set_ylabel('Value')
    ax.tick_params(axis='x', rotation=30)
    ax.grid(True, axis='y', alpha=0.3)

plt.tight_layout()
plt.show()
print('Key insight: GR and RHOB are usually the best separators for lithology')
```

Out [16]:



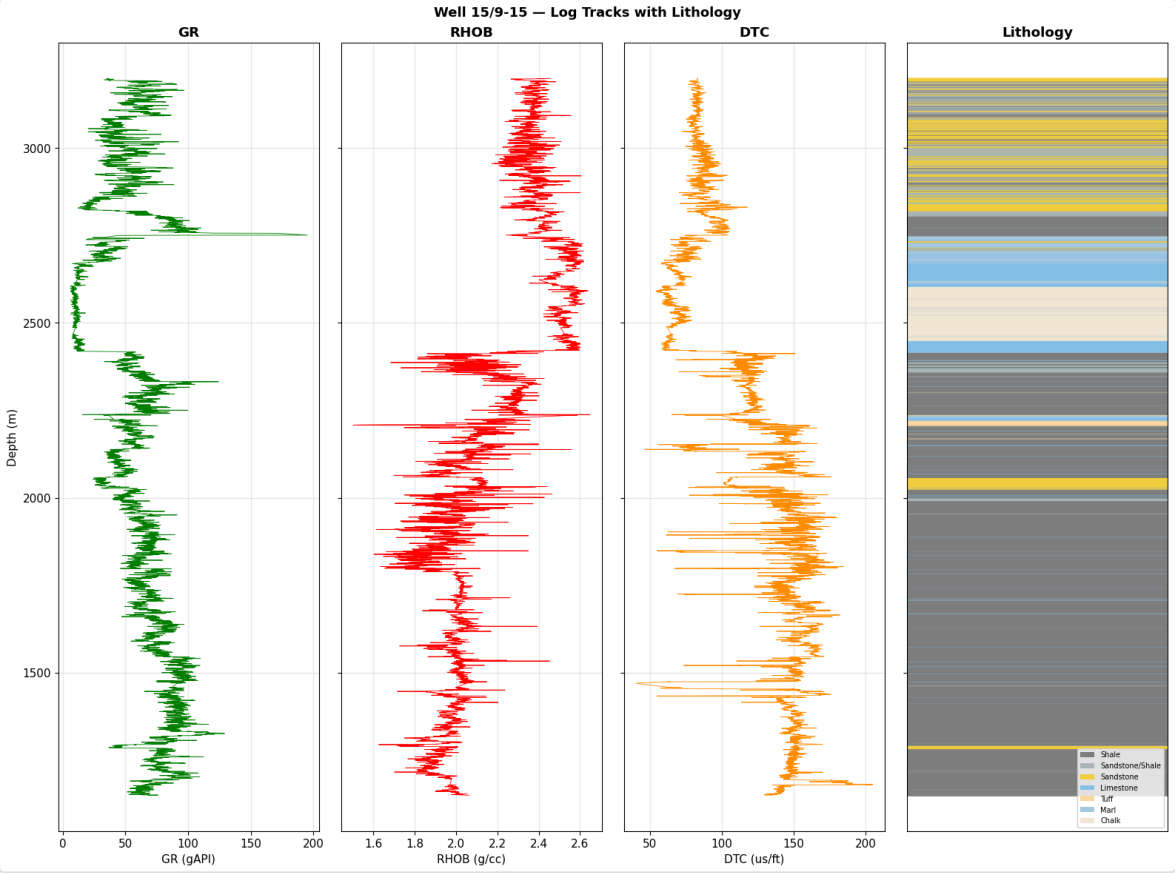

```
Key insight: GR and RHOB are usually the best separators for lithology
```

In [17]:

```python
# ── 5.4 Log Track — one well with lithology color fill ──────────────────
# Pick one well to display as classic log tracks

# Define a color for each lithology class
lith_colors = {
    'Shale':           '#808080',
    'Sandstone':       '#F4D03F',
    'Sandstone/Shale': '#AAB7B8',
    'Limestone':       '#85C1E9',
    'Chalk':           '#F0E6D3',
    'Marl':            '#A9CCE3',
    'Halite':          '#F8C471',
    'Anhydrite':       '#D7BDE2',
    'Dolomite':        '#A3E4D7',
    'Tuff':            '#FAD7A0',
    'Coal':            '#2C3E50',
}

df_plot = df_train[df_train['WELL'] == '15/9-15'].copy()
depth   = df_plot['DEPTH_MD'].values

fig, axes = plt.subplots(1, 4, figsize=(16, 12), sharey=True)
fig.suptitle('Well 15/9-15 — Log Tracks with Lithology', fontsize=13, fontweight='bold')

# Track 1: GR
axes[0].plot(df_plot['GR'], depth, 'green', lw=0.6)
axes[0].set_xlabel('GR (gAPI)'); axes[0].set_title('GR', fontweight='bold')
axes[0].set_ylabel('Depth (m)'); axes[0].invert_yaxis(); axes[0].grid(True, alpha=0.3)

# Track 2: RHOB
axes[1].plot(df_plot['RHOB'], depth, 'red', lw=0.6)
axes[1].set_xlabel('RHOB (g/cc)'); axes[1].set_title('RHOB', fontweight='bold')
axes[1].invert_yaxis(); axes[1].grid(True, alpha=0.3)

# Track 3: DTC
axes[2].plot(df_plot['DTC'], depth, 'darkorange', lw=0.6)
axes[2].set_xlabel('DTC (us/ft)'); axes[2].set_title('DTC', fontweight='bold')
axes[2].invert_yaxis(); axes[2].grid(True, alpha=0.3)

# Track 4: Lithology color bar
ax = axes[3]
ax.set_title('Lithology', fontweight='bold')
# Plot colored blocks for each depth sample
```

```
for j in range(len(df_plot) - 1):
    lith = df_plot[TARGET].iloc[j]
    color = lith_colors.get(lith, 'white')
    d_top = depth[j]
    d_bot = depth[j+1]
    ax.fill_betweenx([d_top, d_bot], 0, 1, color=color, alpha=0.9)

# Legend
from matplotlib.patches import Patch
liths_present = df_plot[TARGET].unique()
legend_elements = [Patch(facecolor=lith_colors.get(l, 'white'), label=l)
                   for l in liths_present]
ax.legend(handles=legend_elements, loc='lower right', fontsize=7)
ax.set_xlim(0, 1); ax.set_xticks([])
ax.invert_yaxis(); ax.grid(False)

plt.tight_layout()
plt.show()
```

Out [17]:



⚙️ STEP 6 — Label Encoding & Feature Scaling

Label encoding: Converts text class names → numbers (required by sklearn)

- Shale → 0, Sandstone → 1, etc.

StandardScaler: Normalizes each feature to mean=0, std=1

In [18]:

```
# ── Extract arrays ──────────────────────────────────────────────
X_train_full = df_train[FEATURES].values
X_blind      = df_blind[FEATURES].values

# ── Encode lithology labels: text → integer ──────────────────────
# LabelEncoder learns the mapping from the training data
le = LabelEncoder()
```

```
le.fit(df_train[TARGET])            # learn class names from training data

y_train_full = le.transform(df_train[TARGET])  # convert train labels
y_blind      = le.transform(df_blind[TARGET])  # convert blind labels (same mapping)

# Print the mapping so we know which number = which lithology
print('Label encoding mapping:')
for i, name in enumerate(le.classes_):
    print(f'  {i:2d} = {name}')

# ── Scale features ──────────────────────────────────────────────
scaler = StandardScaler()
scaler.fit(X_train_full)                        # fit on training data ONLY
X_train_scaled = scaler.transform(X_train_full)
X_blind_scaled = scaler.transform(X_blind)       # same scaler on blind well

print(f'\nTraining set : {X_train_scaled.shape}')
print(f'Blind well   : {X_blind_scaled.shape}')
```

Out [18]:

```
Label encoding mapping:
   0 = Anhydrite
   1 = Chalk
   2 = Coal
   3 = Dolomite
   4 = Limestone
   5 = Marl
   6 = Sandstone
   7 = Sandstone/Shale
   8 = Shale
   9 = Tuff

Training set : (67858, 6)
Blind well   : (11927, 6)
```

🎯 STEP 7 — Train / Validation Split

80% of training data for learning, 20% for checking during model comparison. Blind well (15/9-13) stays completely separate.

In [19]:

```
# ── 80% train / 20% validation ──────────────────────────────────
X_train, X_val, y_train, y_val = train_test_split(
    X_train_scaled,
    y_train_full,
    test_size=0.2,
    random_state=42,
    stratify=y_train_full    # keep same class % in both splits
)

print(f'Train      : {X_train.shape[0]:,} rows')
print(f'Validation : {X_val.shape[0]:,} rows')
print(f'Blind well : {X_blind_scaled.shape[0]:,} rows')
```

Out [19]:

```
Train      : 54,286 rows
Validation : 13,572 rows
Blind well : 11,927 rows
```

🏃 STEP 8 — Model Comparison (LazyPredict Style)

We train 7 classifiers and rank them by **Accuracy** and **F1 Score** on the validation set.

F1 score is better than accuracy when classes are imbalanced (like here — lots of Shale).

In [20]:

```
# ── Define models to compare ────────────────────────────────────
models_to_test = [
    ('Logistic Regression', LogisticRegression(max_iter=1000, random_state=42)),
```

```
    ('Naive Bayes',         GaussianNB()),
    ('K-Nearest Neighbors', KNeighborsClassifier(n_neighbors=5)),
    ('Decision Tree',       DecisionTreeClassifier(random_state=42)),
    ('Random Forest',       RandomForestClassifier(n_estimators=100, random_state=42, n_jobs=-1)),
    ('XG Boost',            XGBClassifier(n_estimators=100, random_state=42, n_jobs=-1)),
    ('Gradient Boosting',   GradientBoostingClassifier(n_estimators=100, random_state=42)),
]

# ── Train each model and record performance ──────────────────────────
results = []

for model_name, model in models_to_test:
    print(f'Training: {model_name}...', end=' ')

    model.fit(X_train, y_train)          # train
    y_pred = model.predict(X_val)        # predict on validation set

    acc = accuracy_score(y_val, y_pred)
    f1  = f1_score(y_val, y_pred, average='weighted')   # weighted = accounts for class imbalance

    results.append({'Model': model_name, 'Accuracy': round(acc, 4),
                    'F1 (weighted)': round(f1, 4), 'object': model})

    print(f'Accuracy={acc:.4f}  F1={f1:.4f}')

print('\nAll models trained!')
```

Out [20]:

```
Training: Logistic Regression... Accuracy=0.7871  F1=0.7584
Training: Naive Bayes... Accuracy=0.6496  F1=0.6414
Training: K-Nearest Neighbors... Accuracy=0.8935  F1=0.8913
Training: Decision Tree... Accuracy=0.8691  F1=0.8689
Training: Random Forest... Accuracy=0.9137  F1=0.9108
Training: XG Boost... Accuracy=0.8999  F1=0.8974
Training: Gradient Boosting... Accuracy=0.8601  F1=0.8535

All models trained!
```

In [21]:

```
# ── Show ranked results ──────────────────────────────────────────────
results_df = pd.DataFrame(results).drop(columns=['object'])
results_df = results_df.sort_values('F1 (weighted)', ascending=False).reset_index(drop=True)
results_df.index += 1

print('MODEL COMPARISON — Ranked by F1 Score (Validation Set)')
display(results_df)

best_model_name = results_df.iloc[0]['Model']
print(f'\nBest model: {best_model_name}')
```

Out [21]:

```
MODEL COMPARISON — Ranked by F1 Score (Validation Set)
```

	Model	Accuracy	F1 (weighted)
1	Random Forest	0.9137	0.9108
2	XG Boost	0.8999	0.8974
3	K-Nearest Neighbors	0.8935	0.8913
4	Decision Tree	0.8691	0.8689
5	Gradient Boosting	0.8601	0.8535
6	Logistic Regression	0.7871	0.7584
7	Naive Bayes	0.6496	0.6414

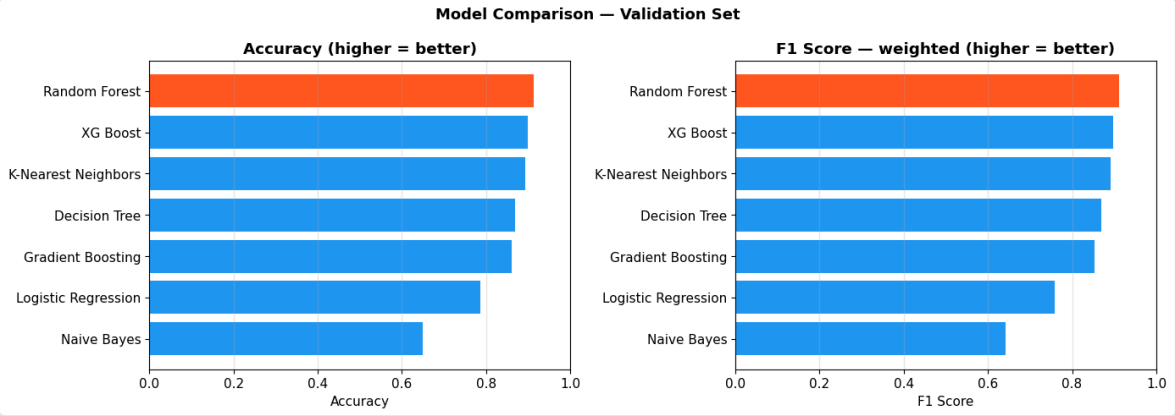
```
Best model: Random Forest
```

In [22]:

```
# ── Bar chart comparison ─────────────────────────────────────────────
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
fig.suptitle('Model Comparison — Validation Set', fontsize=13, fontweight='bold')
```

```
plot_df = results_df.sort_values('F1 (weighted)', ascending=True)
colors  = ['#FF5722' if n == best_model_name else '#2196F3' for n in plot_df['Model']]

axes[0].barh(plot_df['Model'], plot_df['Accuracy'], color=colors)
axes[0].set_title('Accuracy (higher = better)', fontweight='bold')
axes[0].set_xlabel('Accuracy')
axes[0].set_xlim(0, 1)
axes[0].grid(True, axis='x', alpha=0.3)

axes[1].barh(plot_df['Model'], plot_df['F1 (weighted)'], color=colors)
axes[1].set_title('F1 Score — weighted (higher = better)', fontweight='bold')
axes[1].set_xlabel('F1 Score')
axes[1].set_xlim(0, 1)
axes[1].grid(True, axis='x', alpha=0.3)

plt.tight_layout()
plt.show()
print(f'Red bar = {best_model_name} (best model)')
```

Out [22]:




```
Red bar = Random Forest (best model)
```

🔬 STEP 9 — Best Model: Detailed Training

Retrain on all training data, check feature importance and classification report.

In [23]:

```
# ── Retrieve the best model object ──────────────────────────────
best_model = None
for entry in results:
    if entry['Model'] == best_model_name:
        best_model = entry['object']
        break

# ── Retrain on ALL training data (train + val combined) ──────────
X_all = np.vstack([X_train, X_val])
y_all = np.concatenate([y_train, y_val])

print(f'Retraining {best_model_name} on {len(X_all):,} samples...')
best_model.fit(X_all, y_all)
print('Done!')
```

Out [23]:

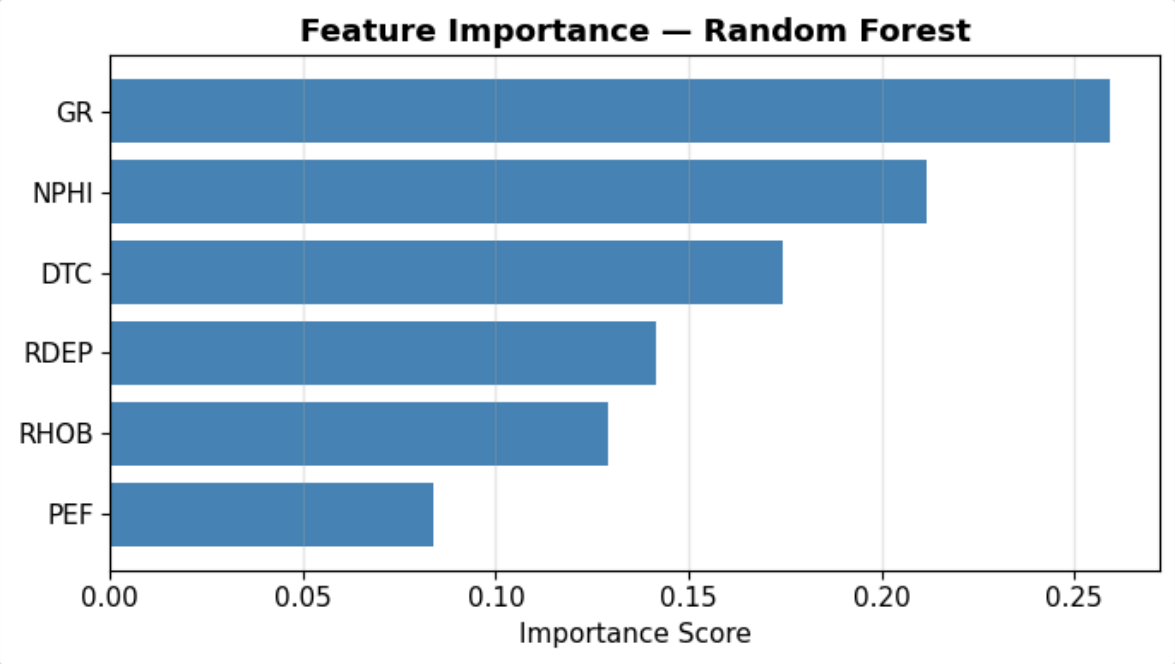
```
Retraining Random Forest on 67,858 samples...
Done!
```

In [24]:

```
# ── Feature Importance (tree-based models only) ──────────────────
if hasattr(best_model, 'feature_importances_'):
    imp_df = pd.DataFrame({'Feature': FEATURES,
                           'Importance': best_model.feature_importances_})
    imp_df = imp_df.sort_values('Importance', ascending=True)
```

```
    fig, ax = plt.subplots(figsize=(7, 4))
    ax.barh(imp_df['Feature'], imp_df['Importance'], color='steelblue')
    ax.set_title(f'Feature Importance — {best_model_name}', fontweight='bold')
    ax.set_xlabel('Importance Score')
    ax.grid(True, axis='x', alpha=0.3)
    plt.tight_layout()
    plt.show()
else:
    print(f'{best_model_name} does not have feature_importances_')
```

Out [24]:



🧪 STEP 10 — Blind Well Prediction & Evaluation

Final test on **well 15/9-13** — the model has never seen this well.

In [25]:

```
# ── Predict lithology on blind well ──────────────────────────────
y_blind_pred = best_model.predict(X_blind_scaled)

# Convert numeric labels back to lithology names for readability
y_blind_names = le.inverse_transform(y_blind)
y_pred_names  = le.inverse_transform(y_blind_pred)

acc_b = accuracy_score(y_blind, y_blind_pred)
f1_b  = f1_score(y_blind, y_blind_pred, average='weighted')

print('=' * 50)
print(f'  BLIND WELL (15/9-13) — Final Results')
print('=' * 50)
print(f'  Model    : {best_model_name}')
print(f'  Accuracy : {acc_b:.4f}  ({acc_b*100:.1f}%)')
print(f'  F1 Score : {f1_b:.4f}')
print('=' * 50)
```

Out [25]:

```
==================================================
  BLIND WELL (16/1-2) — Final Results
==================================================
  Model    : Random Forest
  Accuracy : 0.7910  (79.1%)
  F1 Score : 0.7662
==================================================
```

In [26]:

```
# ── Classification Report ──────────────────────────────────────────────
# Shows precision, recall, and F1 for EACH lithology class separately
# precision = of all samples predicted as Shale, how many were really Shale?
# recall    = of all real Shale samples, how many did the model catch?

print('Classification Report — Blind Well (16/1-2):')
print(classification_report(y_blind_names, y_pred_names, zero_division=0))
```

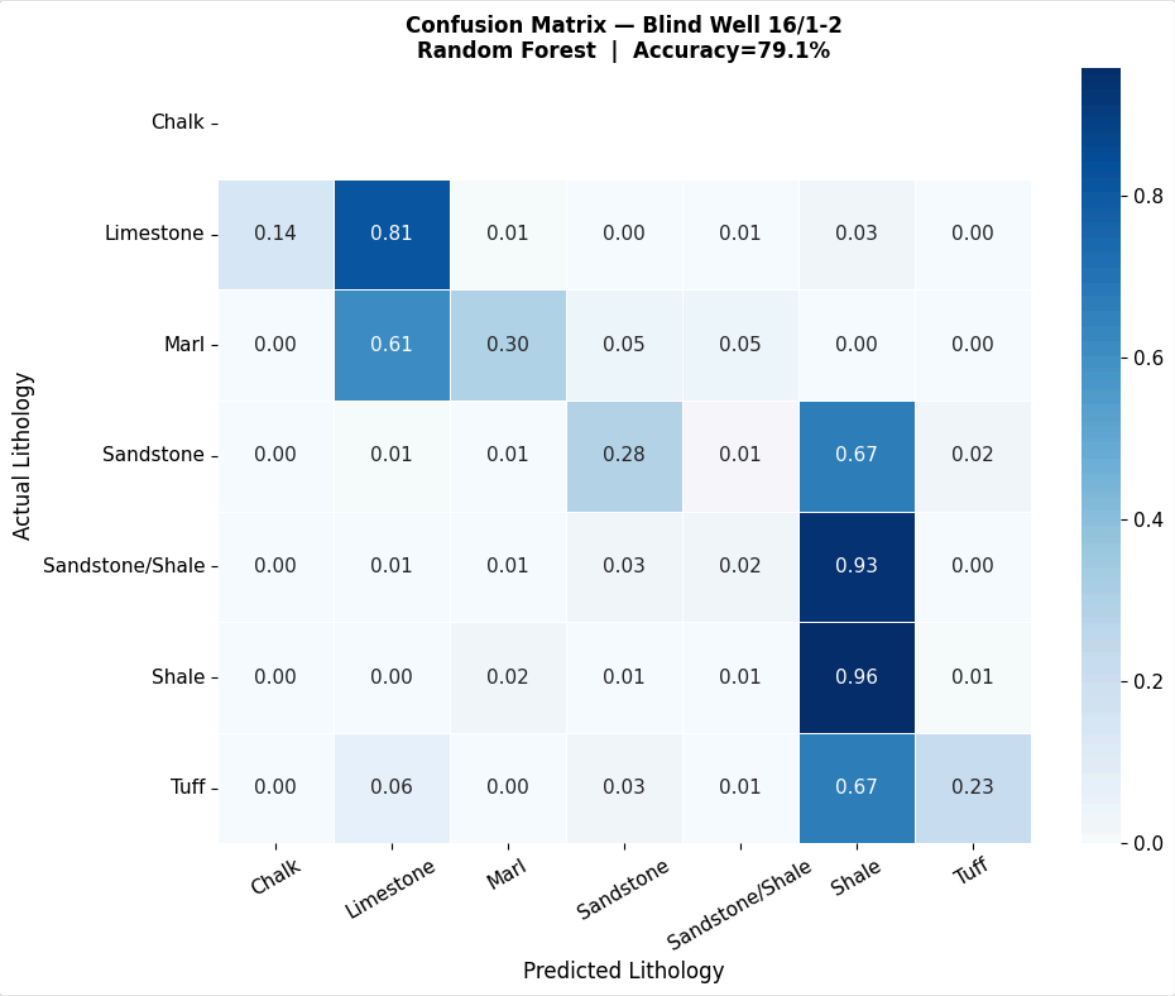
Out [26]:

```
Classification Report — Blind Well (16/1-2):
                 precision    recall  f1-score   support

          Chalk       0.00      0.00      0.00         0
      Limestone       0.78      0.81      0.79      1409
           Marl       0.42      0.30      0.35       419
      Sandstone       0.84      0.28      0.43      1544
Sandstone/Shale       0.07      0.02      0.03       351
          Shale       0.83      0.96      0.89      8031
           Tuff       0.27      0.23      0.25       173

       accuracy                           0.79     11927
      macro avg       0.46      0.37      0.39     11927
   weighted avg       0.78      0.79      0.77     11927
```

In [27]:

```
# ── Confusion Matrix ───────────────────────────────────────────────────
# Rows = actual lithology
# Columns = predicted lithology
# Diagonal = correct predictions
# Off-diagonal = wrong predictions (confusion between classes)

# Get the unique classes present in blind well
present_classes = sorted(set(y_blind_names) | set(y_pred_names))

cm = confusion_matrix(y_blind_names, y_pred_names, labels=present_classes)

fig, ax = plt.subplots(figsize=(10, 8))

# Plot heatmap: normalize by row (shows % of each actual class predicted correctly)
cm_norm = cm.astype(float) / cm.sum(axis=1, keepdims=True)
sns.heatmap(
    cm_norm,
    annot=True,                  # show numbers in each cell
    fmt='.2f',                   # 2 decimal places
    cmap='Blues',                # blue = high values
    xticklabels=present_classes,
    yticklabels=present_classes,
    linewidths=0.5,
    ax=ax
)

ax.set_xlabel('Predicted Lithology', fontsize=12)
ax.set_ylabel('Actual Lithology', fontsize=12)
ax.set_title(f'Confusion Matrix — Blind Well 16/1-2\n{best_model_name}  |  Accuracy={acc_b*100:.1f}%',
             fontsize=12, fontweight='bold')
ax.tick_params(axis='x', rotation=30)
ax.tick_params(axis='y', rotation=0)

plt.tight_layout()
plt.show()

print('Diagonal = correct predictions (darker blue = better)')
print('Off-diagonal = where the model confused one lithology for another')
```

Out [27]:



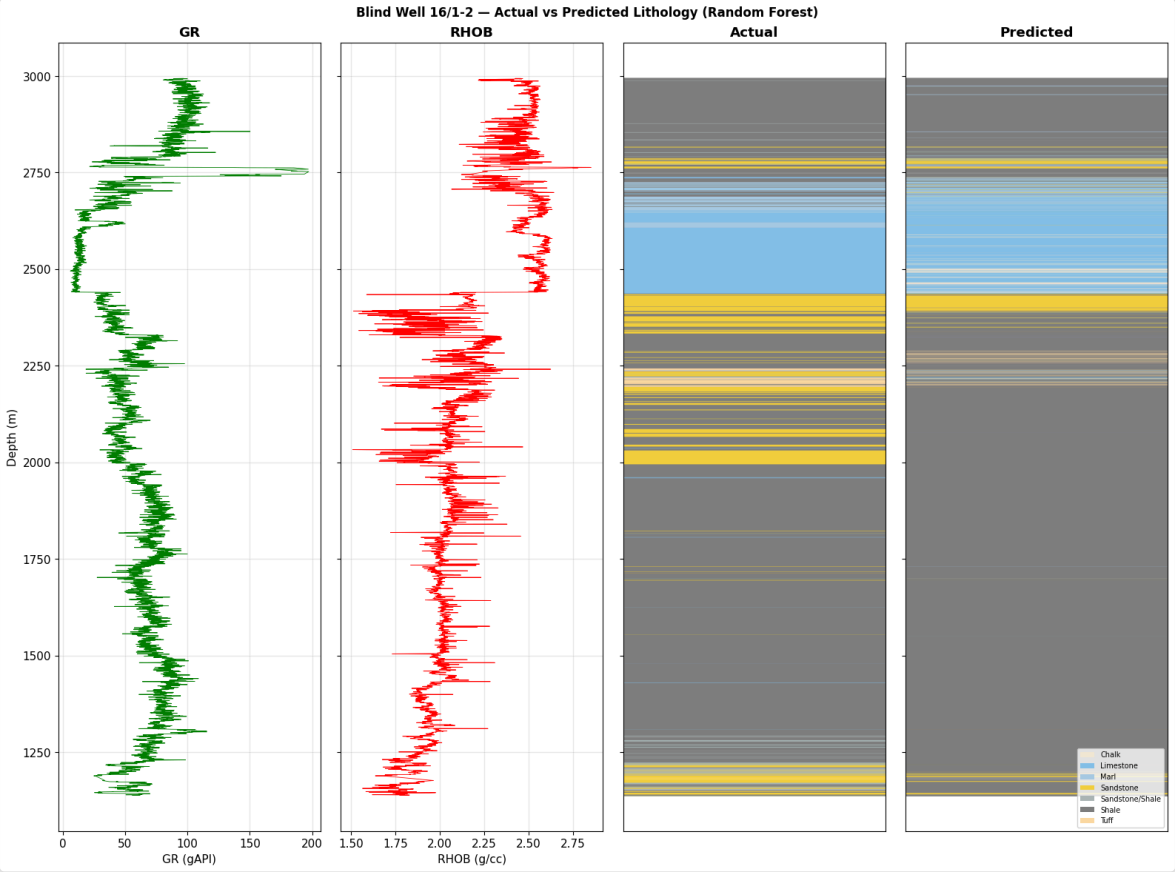

```
Diagonal = correct predictions (darker blue = better)
Off-diagonal = where the model confused one lithology for another
```

In [28]:

```python
# ── Log track: Actual vs Predicted lithology vs depth ──────────────────
depth_blind = df_blind['DEPTH_MD'].values

fig, axes = plt.subplots(1, 4, figsize=(16, 12), sharey=True)
fig.suptitle(f'Blind Well 16/1-2 — Actual vs Predicted Lithology ({best_model_name})',
             fontsize=12, fontweight='bold')

# Track 1: GR
axes[0].plot(df_blind['GR'], depth_blind, 'green', lw=0.6)
axes[0].set_xlabel('GR (gAPI)'); axes[0].set_title('GR', fontweight='bold')
axes[0].set_ylabel('Depth (m)'); axes[0].invert_yaxis(); axes[0].grid(True, alpha=0.3)

# Track 2: RHOB
axes[1].plot(df_blind['RHOB'], depth_blind, 'red', lw=0.6)
axes[1].set_xlabel('RHOB (g/cc)'); axes[1].set_title('RHOB', fontweight='bold')
axes[1].invert_yaxis(); axes[1].grid(True, alpha=0.3)

# Track 3: Actual Lithology
ax = axes[2]
ax.set_title('Actual', fontweight='bold')
for j in range(len(df_blind) - 1):
    lith  = y_blind_names[j]
    color = lith_colors.get(lith, 'white')
    ax.fill_betweenx([depth_blind[j], depth_blind[j+1]], 0, 1, color=color, alpha=0.9)
ax.set_xlim(0, 1); ax.set_xticks([])
ax.invert_yaxis(); ax.grid(False)

# Track 4: Predicted Lithology
ax = axes[3]
ax.set_title('Predicted', fontweight='bold')
```

```
for j in range(len(df_blind) - 1):
    lith  = y_pred_names[j]
    color = lith_colors.get(lith, 'white')
    ax.fill_betweenx([depth_blind[j], depth_blind[j+1]], 0, 1, color=color, alpha=0.9)

# Shared legend
from matplotlib.patches import Patch
all_liths = sorted(set(y_blind_names) | set(y_pred_names))
legend_elements = [Patch(facecolor=lith_colors.get(l, 'white'), label=l) for l in all_liths]
ax.legend(handles=legend_elements, loc='lower right', fontsize=7)
ax.set_xlim(0, 1); ax.set_xticks([])
ax.invert_yaxis(); ax.grid(False)

plt.tight_layout()
plt.show()
print('Left color column = actual lithology | Right = what the model predicted')
```

Out [28]:




```
Left color column = actual lithology | Right = what the model predicted
```

📊 STEP 11 — Final Summary

In [29]:

```
# ── Print full summary ──────────────────────────────────────────
print('=' * 60)
print('          SESSION SUMMARY — LITHOLOGY CLASSIFICATION')
print('=' * 60)
print(f'  Features       : {FEATURES}')
print(f'  Target         : LITH ({len(le.classes_)} classes)')
print(f'  Classes        : {list(le.classes_)}')
print(f'  Blind well     : {BLIND_WELL}')
print(f'  Training rows  : {len(X_all):,}')
print(f'  Blind rows     : {len(X_blind_scaled):,}')
print('\n  ALL MODELS RANKED:')
display(results_df)
print(f'\n  BEST MODEL ON BLIND WELL:')
```

```python
print(f'  Model    : {best_model_name}')
print(f'  Accuracy : {acc_b*100:.1f}%')
print(f'  F1 Score : {f1_b:.4f}')
print('=' * 60)
```

Out [29]:

```
============================================================
          SESSION SUMMARY — LITHOLOGY CLASSIFICATION
============================================================
  Features       : ['GR', 'RHOB', 'NPHI', 'PEF', 'RDEP', 'DTC']
  Target         : LITH (10 classes)
  Classes        : ['Anhydrite', 'Chalk', 'Coal', 'Dolomite', 'Limestone', 'Marl', 'Sandstone', 'Sandstone/Shale', 'Shale', 'Tuff']
  Blind well     : 15/9-13
  Training rows  : 67,858
  Blind rows     : 11,927

  ALL MODELS RANKED:
```

	Model	Accuracy	F1 (weighted)
1	Random Forest	0.9137	0.9108
2	XG Boost	0.8999	0.8974
3	K-Nearest Neighbors	0.8935	0.8913
4	Decision Tree	0.8691	0.8689
5	Gradient Boosting	0.8601	0.8535
6	Logistic Regression	0.7871	0.7584
7	Naive Bayes	0.6496	0.6414

```
  BEST MODEL ON BLIND WELL:
  Model    : Random Forest
  Accuracy : 79.1%
  F1 Score : 0.7662
============================================================
```

Key Takeaways

Concept	What we learned
Classification vs Regression	Here we predict a **category** (rock type), not a number
Class imbalance	Shale dominates — F1 score is fairer than accuracy
Label Encoding	Converts text labels → numbers for sklearn
Stratified split	Keeps same class % in train and val splits
Confusion matrix	Shows which lithologies the model confuses with each other
Precision	Of all predicted Shale → how many really are Shale?
Recall	Of all real Shale → how many did the model find?
F1 Score	Balance between precision and recall
Blind well track	Visual comparison of actual vs predicted lithology by depth